Coelacanth: A Cross-Field Rediscovery Benchmark for Precedent Checkers, and What Closed-Book Recall Does to It

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Abstract

Three Claude models with no tools recall nine of the ten documented cross-field rediscoveries in Coelacanth v0, so a retrieval score on them cannot be separated from memory. Six are recalled through a quiz-gold cross-field work and three only through works the models named, which an annotator rated and the author adjudicated. The tenth would be a closed-book failure only if Opus 5.5's declines were ignored and four undecided Haiku 4.5 loose matches read as misses. Coelacanth v0 measures how often precedent checkers find earlier work stated in another field's vocabulary. Its 60 items, 49 of them adjudicated, also hold attributed controls, AI-generated methods and 15 negatives. A system sees only a field-native claim and its date; each claim-candidate pair carries one of six typed relations. Five paid annotators labelled 136 pairs twice, starting from three options a model proposed, and the author adjudicated; agreement is below the thresholds the protocol fixed before any label. Only two Claude Code web-search agent configurations are run, with and without a prior-art skill. They find a cross-field gold work in the top 10 for 3 and 5 of the ten, and Sonnet 5.5 without tools names one for 6 (4 on quiz gold alone). Both agents call almost every negative related prior work. We reproduce the fingerprint rows of a published twin-retrieval baseline exactly. Code (MIT): https://github.com/IskakovDamir/coelacanth-benchmark; data (CC-BY-4.0, encrypted, password in the datasheet): https://doi.org/10.17605/OSF.IO/672RJ.

Keywords: cross-field rediscovery; novelty checking; benchmark; memorization; closed-book probing.

1 Introduction

In 1994 a paper in *Diabetes Care* proposed a model for the area under glucose tolerance curves [26]; a letter the same year was titled "Tai's formula is the trapezoidal rule" [17]. Its wording contains no term from numerical analysis, so a search in its own vocabulary finds nothing to match; we use it as worked example E1, excluded from the benchmark.

Several tools now answer some form of "has anyone done this before?" [2, 4, 9, 11, 23, 31]. We found no published measurement of how often any of them retrieves earlier work stated in another field's vocabulary (checked again on 12 October 2026).

Existing novelty benchmarks cannot supply it: they take ground truth from reviewers, citations or publication order, or use no labels [15, 16, 22, 28, 30], so a cross-field twin that shares no citation with the claim cannot enter them. The closest resource, Kulikowski's 210 cross-field twin pairs [14], has no verdict labels or memorization control.

Memorization is the second difficulty: a model may name the earlier work from memory, and in v0 it nearly always does (Section 5), so a benchmark for these tools needs closed-book recall per item and results that report recalled and unrecalled items separately.

The benchmark takes its name from the coelacanth, a fish known only from fossils and thought extinct since the end of the Cretaceous, about 66 million years ago, until a living one was caught off South Africa in 1938. This paper describes Coelacanth v0 and what it measures:

- **(C1) The resource:** field-native claims, typed pair labels from paid annotators and the author's adjudication, claim charts with verbatim-checked quote pairs, and closed-book recall per item, model and prompt (Section 3).
- **(C2) Closed-book recall:** 9 of the 10 documented rediscoveries in `R` are recalled by some model, 6 through a quiz-gold cross-field work and 3 only through works the models named (Section 5).
- **(C3) A pilot measurement** of two Claude Code agent configurations, with and without a public prior-art skill, on `R` (Section 7).
- **(C4) An exact reproduction** of Kulikowski's fingerprint TF-IDF rows and a closed-pool protocol (Section 6).
- **(C5) The release:** code and datasheet in a public repository, data in an encrypted archive (Section 10).

2 Related Work

Table 1 sets the benchmark beside the closest resources and checkers.

Twin retrieval. Kulikowski strips field vocabulary into mechanism skeletons and retrieves cross-field twins with TF-IDF (109 papers, 5,812 cross-field pairs, 210 twins) [14]; we reproduce his numbers (Section 6) and add typed labels, open search and closed-book recall.

Checkers. OpenNovelty requires verbatim quote pairs for every refutation [31]; Ai2's Idea Novelty Checker compares the facets of an idea with retrieved papers (accuracy 0.81 on 32 expert-labelled ideas) [23] and scores 8 to 29 macro-F1 points below a cheap prompted judge on ICLR 2026 pairs [25]. Theorizer [9], Hypothetica (abstract only) [27], FutureHouse's Owl [4], Idea-Catalyst [11], SciAtlas [19] and the prior-art skill [2] label, retrieve or translate across fields, and none reports a cross-field retrieval rate.

Benchmarks. AViD returns equivalent, generalization, specialization or different for Lean-formalised mathematics [18]; we adopt its types. NovGauge, NovBench, RINoBench, RQ-Bench, SchNovel and an axiomatic benchmark of novelty metrics take ground truth from reviewer overlap, citations, publication order or reference-pool manipulation [15,16,22,24,28,30]; none labels cross-field rediscoveries.

Rediscovery and contamination. Kim, Kojaku and Ahn detect same-year simultaneous discoveries from citation graphs [12]; the Gradient Coupling Inquiry catalogues 6 independent rediscoveries among 18 cross-domain instances [7]; within one field, 45.3% of agent-designed ML methods share an algorithmic coordinate with a human method [29] and 24% of 50 AI-generated research documents were paraphrased or borrowed [6]. A naming probe can miss contamination that moves performance [20], guided instructions detect more [5], and accuracy tracks pretraining frequency [10]. Sainz et al. ask for contamination to be measured per benchmark [21], and Jacovi et al. recommend releasing test data encrypted [8].

Table 1. Related resources and checkers by property. Tags: [V] read on the primary page or full text; [S] secondary or abstract only. "Not reported" means the source reports nothing on the property, under the coverage of our review.

Resource or checker	Ground truth for prior work	Typed labels	Closed-book control	Non-comp. items	Verbatim evidence
Kulikowski 2026 [V]	210 curated twins	no (ranked pairs)	not reported	excluded	no (human check)
OpenNovelty [V]	none quantitative	refute / cannot refute / unclear	not reported	untested	quote pairs checked in code
Ai2 INC [V]	32 expert-labelled ideas	novel / not novel	not reported	untested	no
Ai2 Theorizer [V]	label proportions only	established / derivable / new	not reported	any S2 field	not documented [S]
Owl / Precedent [V]	no published accuracy	free text, found flag	not reported	biology-first	no (LLM summaries)
Idea-Catalyst [V]	LLM-judge win rates	coverage levels	not reported	as sources only	no
AViD [V]	26 within-field withdrawals	four equivalence types	not reported	no (mathematics)	no
NovGauge [V abstract]	reviewer overlap, co-citation	task / problem / method	not reported	no	no
Gradient Coupling [V]	18 instances, 6 independent	rediscovery or bridge code	no	physics of durability	no
Coelacanth v0	documented cross-field rediscoveries, labelled by annotators, adjudicated by the author	six pair labels, item verdict	two probes, three models	yes	claim chart, quotes checked in code

3 Benchmark Construction

3.1 Sources

Declarations. Letters, comments, errata and withdrawal notes that state an equivalence name both works [18]. Two OpenAlex phrase searches returned 3,172 and 5,591 works, and Claude Haiku 4.5 screened them. Each sampled or screened title had two blind Claude readings: the sample projects about 20 usable cross-field declarations (95% interval 8.5 to 43), and from these readings Claude chose the 13 declarations of b01 and b02 under the author's delegation. The annotators' labels decide which of them have gold. **Kulikowski's twins.** Of his papers, 72 have an earlier cross-field twin. Fifty-four name the method and are attributed imports; 8 of them, drawn with a seed, form the control subset `C`. The 14 that name it partly or not at all are the rediscovery positives, and 4 are unmarked. **Other sources.** About 4 of the Gradient Coupling catalogue's 6 independent rediscoveries are usable, and two non-computational controls, group 2 innate lymphoid cells and RANKL, each have a documented unification under several names. Ten of 54 methods that Claude proposed as new form the descriptive stratum `A`, and the negatives are Kulikowski's topical distractors.

Composition. Batch b01 holds 30 items with 87 candidates and 161 quote pairs, and batch b02 30 items, among them the 10 AI-origin items, with 49 candidates and 111 quote pairs. A third batch, b03 (cf-0068 and cf-0069, 3 pairs), was labelled and probed but is left out of v0's counts. Table 2 gives the composition after labelling.

`N`, the items whose adjudicated verdict is `nothing_close`, holds 15: 9 near-negatives and 6 mined positives that came out empty; the tenth near-negative, cf-0064, has gold. cf-0004 has no

Table 2. Benchmark v0, batches b01 and b02. "With gold": at least one candidate rated `similar_mechanism` or stronger (any relation but `different`) by the quiz and the author's adjudication. `R`, recalled and `H` are defined in Section 3.5.

Group	Source	Items	Computational	With gold	`R`	Recalled	`H`
positive	declaration	13	12	11	4	4	0
positive	Kulikowski	14	14	10	6	5	0
positive	catalogue	2	2	2	0	0	0
positive	post-cutoff lead	1	1	1	0	0	0
positive	non-computational control	2	0	1	0	0	0
attributed control (`C`)	Kulikowski	8	8	8	0	0	0
near-negative	Kulikowski	10	9	1	0	0	0
AI-origin (`A`)	generated	10	not counted	no candidates yet	0	0	0
Total		60		34	10	9	0

adjudicated label because the author excluded its only candidate, a book readable only in its chapter titles. cf-0044 is in neither the recalled nor the `H` column because Opus 5.5 declined every sample (Section 5.3).

3.2 Field-native claims

Systems see a claim text and its date and nothing else. The text keeps the source field's own vocabulary, removes author names, titles, years, venues and the coined method name, and has at most 120 words; a script checks it. Claude drafted every claim text as pre-fill; annotators checked 51 of them, and 21 of their fixes were kept (the 9 near-negatives are unchecked).

3.3 Claim charts and evidence

Claude drafted each claim's chart of two to six elements and the quote pairs (at most 40 words per side) that show an element covered by a candidate; every quote pair is checked verbatim against the cached texts after normalisation. Of the 135 candidates kept in v0 (cf-0004's was excluded), Claude proposed 33, 20 of them gold, from the gold lists it chose for the pilots (Section 5.1). Coverage is partial: 35 candidates have no quote pair, as do 19 of the 70 gold pairs. The quiz collected no key flags, confidence or element coverage. The loader's validation therefore flags every frozen item, because no chart has a key element, and also the 64 `partial` and 3 `specialization` labels, which lack the coverage they require. The labels rest on the annotators' judgement of the pre-filled packet.

3.4 Typed labels

Each claim-candidate pair gets one of six relations (Table 3). The item verdict is the strongest relation among the gold candidates, or `nothing_close`, a statement about the search that carries a coverage note. `equivalent` and `specialization` are the "already done" verdicts, whose precision is reported separately because a false "already done" is the costliest error. The labelling guide has ordered decision rules, 14 worked examples (all excluded) and nine clarifications fixed before any human label; when unsure between two labels, the annotator takes the one that claims less prior art.

Table 3. Pair labels. C is the claim, P the earlier candidate. The first three labels carry AViD's names, `similar_mechanism` and `different` map to its different, and `partial` has no counterpart.

Label	Definition (abridged)	Already done
`equivalent`	P states the same mechanism or result as C under the same hypotheses, up to renaming and routine reformulation; every key element covered with the same meaning	yes
`specialization`	C is a strict particular case of P: restricting P yields C with the same meaning	yes
`generalization`	C is strictly stronger; P is obtained from C by fixing a parameter or adding a hypothesis	no
`partial`	P covers at least two key elements of C with the same meaning and leaves at least one uncovered	no
`similar_mechanism`	same functional mechanism or form in a different system or with a different meaning of the variables; every key element has a counterpart, at least one by role only	no
`different`	below `similar_mechanism`	no

3.5 Memorization control

Each positive is probed without tools on Claude Opus 5.5, Sonnet 5.5 and Haiku 4.5, on the field-native text and date, under two prompts. A frozen prompt asks for at most five earliest publications that state the idea, with five samples per model, and a cross-field prompt asks whether the idea appeared earlier in a different field, with three (Section 5.1). Declined samples are retried once, never counted as misses; a person decides borderline matches blind to model and prompt.

Notation. b01 to b03: annotation batches; a1 to a3: annotator IDs; `C`, `A` and `N`: Section 3.1. `R`: the adjudicated rediscovery positives outside `C` and `A` with a person-confirmed cross-field gold work that the later work does not cite. *Recalled*: some sample of some model under either prompt names such a work, or a work the author judged cross-field gold. `H`: the `R` items not recalled, less any item for which some probed model has no valid sample after the retry (protocol decision 6). `H-any`: the `H` items on which no sample names any gold work. `H-post`: the `H` items dated after the training cutoff of every model the compared systems call.

3.6 Leakage

Anything a system retrieves that is dated on or after the claim date is a leakage hit, web pages included (dated by the version read); a leakage hit never counts as a find, and the leakage rate is reported per system. A publication-date filter would have hidden 1 of 107 pre-labelling works checked in OpenAlex (156 work-claim pairs): a 1970 book dated 2004 there, for a 1984 claim, later labelled `different`.

3.7 Annotation: what people decided

The annotators and the author made every relation, cross-field and attribution decision in v0, and Table 4 counts their answers; the annotators decided 4 attribution pairs only after the freeze (below). In the quiz, five annotators worked under three IDs: two people shared a1, two a2 and one used a3, and no person used two IDs. Which of the two people behind a1 or a2 gave an answer is not recorded.

Table 4. What people decided, by task, counted in recorded answers from the repository files.

Task	Decided by	Answers	Covers
Pair label (relation, cross-field answer)	paid annotators	272	136 pairs, two readings each
Disagreements settled	the author	51	one-line reason each
Attribution of cross-field gold pairs	paid annotators	48	24 pairs on 12 `R`-eligible items, two readings each; 4 pairs read as cannot check
Attribution after the freeze, 21 October	paid annotators, jointly (author's account)	4	the 4 cannot-check pairs
Claim-text check	paid annotators	51	the 51 probed items
Pair label, batch b03	paid annotators	6	3 pairs, outside v0's counts
Works named closed-book	paid annotators	31	3 items, one reading each
Named works rated gold	the author	14	14 of the 31 works
Borderline closed-book matches	paid annotators (author's account)	213	first run
Re-decision of borderline matches	the author	30	repeats 30 entries; all agree
Total		**720**	690 distinct; b03 included

Five paid annotators with graduate STEM training (Section 9) labelled the 136 claim-candidate pairs of b01 and b02 in a quiz: for each pair they chose the relation from three options Claude proposed from the pre-filled packet, in fixed precedence order, or any other relation, and said whether the earlier work is from another field. Every pair went to two IDs, so two different people read it; the page never showed which option Claude ranked first. They worked on 10 October 2026 in the author's presence, which kept them from consulting language models. They pasted in comments written outside the quiz page, which we do not use as evidence of reasoning, and spent a median of 12 to 14.5 s of active page time per question.

On the first readings, Krippendorff's alpha [13] (Figure 1) is 0.542 on the six-way relation (95% bootstrap interval 0.396 to 0.688), 0.783 on gold, 0.321 on "already done", which rests on 6 of 272 readings, and 0.794 on cross-field status. All three miss the thresholds of protocol decision 18, 0.667 for the six-way relation and 0.80 for gold and "already done"; the guide revision and second measurement on fresh items that the decision then requires were not done, since no annotators remain. By ID pair the six-way alpha is 0.351 for a1 and a2 (90 pairs), 0.936 for a1 and a3 (28) and 1.0 for a2 and a3 (18). Model-proposed options can anchor annotators [1, 3], and 189 of the 272 answers, 0.695 against 0.333 by chance, equal the option Claude ranked first. None of the 272 answers left the three options.

The quiz did not cover claim texts, key elements or attribution. In two further rounds in the author's presence, the annotators checked the claim texts (Section 3.2) and the attribution of the 24 cross-field gold pairs on the 12 items that could enter `R`; the other 12 of the 36 cross-field gold pairs are on the attributed controls. Codex, a model, repackaged and anonymised their returns at the author's request. The two attribution readings are word for word identical, so no agreement is reported. Both readings marked 4 pairs as impossible to check: three on cf-0012, whose place in `R` rests on them, and one on cf-0031. The annotators decided these 4 on 21 October, after the freeze, from the reference lists the publishers deposited in Crossref, on a page that did not show the earlier decision; before the freeze Claude had decided them from the same lists (Section 9). Three annotators answered together in the author's presence and gave the same answer on every pair,

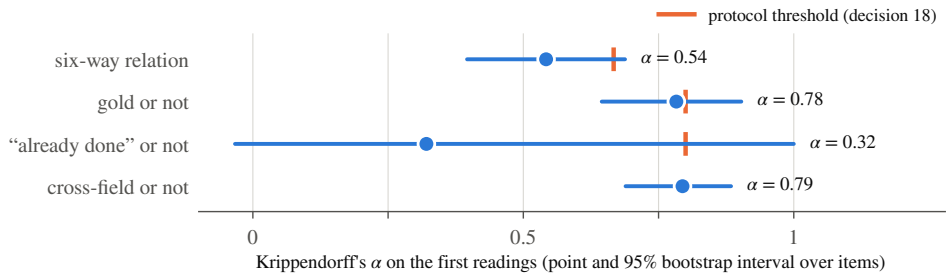


Figure 1. Agreement of the quiz on the 136 pairs of b01 and b02. Points are Krippendorff's alpha on the first readings and bars the 95% percentile bootstrap intervals over items (2,000 resamples); orange ticks are the thresholds of protocol decision 18 (none is set for cross-field status).

which the author recorded once, under ID a1; as one joint reading it has no agreement figure. On all 4 they answered that the later work does not cite the earlier one, so cf-0012 stays in `R`, now on the annotators' reading, and cf-0031, whose attribution was unknown at the freeze, would join it, a logged deviation (Section 5.3). The annotators also labelled batch b03 and judged the 31 named works by hand on printed sheets, which Codex transcribed.

4 Measurement A: Design

Systems and scoring. Two agent configurations are scored: a single agent and the same agent with the public prior-art skill [2], both Claude Code with Sonnet 5.5, with web search and fetch as their only tools and the same caps ($0.40 per item). The single agent also runs on a domain-stripped skeleton of each claim. The memory reference, Sonnet 5.5 without tools under the frozen closed-book prompt, is scored on `R` like a system. Kulikowski's fingerprint is reproduced on its own pool only (Section 6). The other planned systems are not run: the remaining API budget did not cover them and no annotator remains to judge their pools. Pooled judging is replaced by scoring against the adjudicated gold. A retrieved work outside it counts as unjudged, so retrieval figures are lower bounds. The primary metric is cross-field success@10, and since `R` has 10 items, under the 20 that decision 23 requires of a confirmatory test, every comparison is descriptive (Appendix C).

Freeze. On 6 October the protocol fixed the decline rule (decision 6), the rule that subsets under 20 items are only described (decision 23) and agreement thresholds for full independent double labelling (decision 18); on 9 October it changed how `H` is reported (Section 5.2). The three-option quiz that replaced that labelling entered the protocol on 11 October. The rules freeze was planned before labelling and came after the quiz (10 October), the closed-book probes (10 and 11 October) and the adjudication of the named works (11 October), at the tag `protocol-v1.0` on 11 October, 20:33 (UTC+5). The agent configurations were hashed then, and neither agent ran on a benchmark item before the freeze; the memory reference reuses closed-book answers collected before it. Later changes are logged as deviations, among them the skeleton run, decided after the first scored runs had been looked at, and the correction of the gold (Section 7). The SHA-256 of the protocol at the tag (commit 658b3fb of the private development repository) is `06331185afb8be89 1801f5431987ffcde2e40d25c0ba20ea60ecf58ab3867a88`, and the SHA-256 of `splits-v1.json` is `8527d1db9ebfd51559ba10709c6db22d419ec5ea2606b1ee0dbc8157b880637c`; `eval/protocol-v1.0.md` is the protocol exactly as tagged, and `eval/protocol.md` adds the logged deviations.

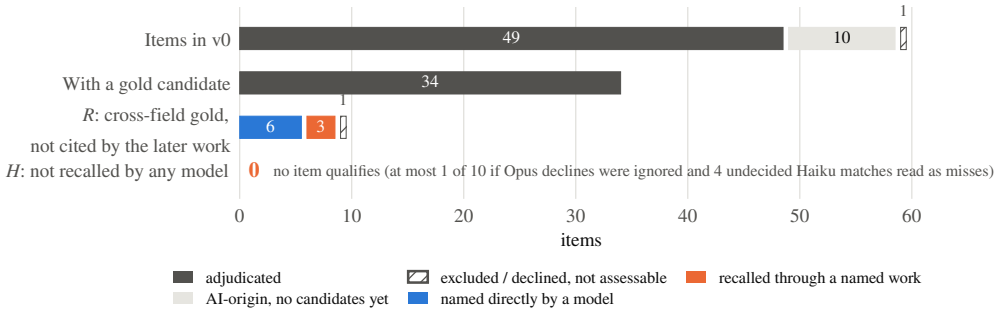


Figure 2. From the items of v0 to `H`. Of the 10 items of `R`, 6 are recalled through quiz gold, 3 only through works the models named, and 1 is not assessable because Opus 5.5 declined every sample; `H` is empty.

5 Closed-Book Recall of Documented Cross-Field Rediscoveries

A published equivalence declaration puts the link itself into the literature the models learn from, so documented rediscoveries are exposed by construction, and a retrieval score on them can measure memory.

5.1 Pilots

Three pilots on Claude-drafted claim texts and gold lists preceded the v0 runs, which replace them (Appendix A). They gave the rule used here: the cross-field prompt caught 8 of the 9 pilot items that the frozen prompt missed, and under either prompt all 16 Kulikowski items were named. What the cross-field prompt surfaces is mostly the method's origin in another field (for example statistics for population pharmacokinetics, linear prediction theory for kriging).

5.2 Supply and what it implies for `H`

The pilots projected `H` at about 5 to 7 items, under the 20 that a test needs, so on 9 Oct, before any human label or system run, the protocol was updated: the union rule stays and `H` is reported first, as k of n, descriptively. Protocol decision 1 (6 October) makes every work a model names closed-book a candidate for its item, judged before `H` is computed; the option rejected on 9 October was a rule redefining gold from what the models named. No subset is called contamination-free, and recall differences between sources, which are confounded with popularity, are not read as a memorization effect.

5.3 v0 runs

All 51 positives of b01 and b02 were probed under both prompts on 10 and 11 October 2026: 1,215 of the 1,224 planned samples are done (1,183 valid, 32 Opus 5.5 samples declined twice) in 1,501 calls for $21.43, with 197 answers reused from the declarations pilot. The 9 missing are Opus 5.5 cross-field samples on three AI-origin items, declined once and not retried; b03's 48 samples ($0.91) are left out. Opus 5.5 declined 75 of its 454 counted answers on b01 and b02, almost all on biology, medicine and immunology claims, and gave no valid sample on five items, which therefore cannot enter `H`: the control cf-0006, the Kulikowski twin cf-0044 and three AI-origin items. Sonnet 5.5 and Haiku 4.5 declined none. The paid annotators decided the 213 borderline matches of the first run by a fixed rubric (the author's account; Appendix B gives their provenance and the remaining bookkeeping, which changes no `R` item's status).

The frozen `R` holds 10 items, without cf-0031, whose attribution was unknown at the freeze (Section 3.7; Figure 2). Six are recalled because a sample names a quiz-gold cross-field work; for cf-0015 this rests only on borderline decisions, all naming Yule 1925 by title. For cf-0007 and cf-0015 the cross-field gold work the models name is one Claude proposed as a candidate (Section 3.3). For three more Kulikowski twins, cf-0013, cf-0036 and cf-0046, no model names a cross-field gold work, and they count as recalled only through works the models named (below). The tenth, cf-0044, is in neither group: Opus 5.5 declined every sample, and Sonnet 5.5 and Haiku 4.5 named no gold in a decided match. Its 5 named works were not judged. If the declines were ignored, Haiku 4.5's four undecided loose matches on it read as misses and none of the 5 works rated gold, `H` would be 1 of 10 (Wilson 0.02 to 0.40). The cross-field prompt raises the number of `R` items whose cross-field gold a model names directly from 4 to 6 for Opus 5.5 and Sonnet 5.5 and from 2 to 4 for Haiku 4.5 (Figure 4, left).

For cf-0013, cf-0036 and cf-0046 the models named 31 earlier works outside the gold lists; an annotator judged each once, under ID a1 or a2, and rated 14 gold, 6 of them cross-field gold. The author adjudicated all 14 on 11 October, keeping 13 readings and lowering one from `equivalent` to `partial`, and confirmed every cross-field reading. Each of the three items therefore counts as recalled, and `H` is empty: 0 of 10 (Wilson 95% interval 0.00 to 0.28), as are `H-any` and `H-post`. One sensitivity line was added after the freeze: the PageRank papers that make cf-0036 recalled are cited by its source, so if recall required an uncited work, `H` would be 1 of 10 (Wilson 0.02 to 0.40). Under the annotators' reading of 21 October (Section 3.7), `R` would hold 11 items, 10 recalled, since Opus 5.5 and Sonnet 5.5 name cf-0031's cross-field gold work under both prompts; `H` would be 0 of 11 (Wilson 0.00 to 0.26), and the frozen 10 stay primary.

6 Kulikowski Reproduction and the Closed-Pool Protocol

We reproduced the TF-IDF rows of Kulikowski's benchmark from his package v1.0.0 with one command (2,181 files verified in about 40 seconds, no model call). The package has two TF-IDF settings: a headline default (unigrams and bigrams, sublinear tf, English stop words) behind every AP in his paper, and scikit-learn's plain unigram vectorizer behind his whole-fingerprint P@1. Every fingerprint number he prints reproduces exactly: under the headline setting, whole-fingerprint AP 0.557 [0.407, 0.720] and skeleton AP 0.513, P@1 0.815 (75 of 92) and MRR 0.860; under the unigram setting, whole-fingerprint P@1 0.826, where AP is 0.441, a value he does not print (Appendix D). The abstract rows differ by at most one query because 5 abstracts are not redistributable. The pool is closed, so a system that searches open literature is penalised for correct earlier works outside the 109 papers; the closed-pool protocol has every system rank only the pool's cross-field candidates for each query, scored by P@1, MRR and AP.

7 Results of Measurement A

Gold is the quiz-labelled gold of the frozen items plus the 14 named works rated gold (Section 5.3), adjudicated before either agent ran. The first scoring used the frozen items' gold alone (Table 5, "Quiz gold"). Every success the named works add is on cf-0046 or on cf-0036 (Section 5.3). The memory reference reuses Sonnet 5.5 frozen-prompt samples, part of the answers those works were drawn from.

Every agent success (Figure 3) is on an item for which the memory reference also names a gold work (Figure 4, right). The prior-art skill finds 2 items that the single agent does not and none the other way round, an exact McNemar p of 0.5 on discordant counts of 2 and 0. The skeleton

Table 5. Retrieval on `R`, text condition unless stated. Success: cross-field success@10, the number of the 10 items with a cross-field gold work in the cleaned top 10; the miss rate has a Wilson 95% interval. "Quiz gold": the same without the 14 named works. The memory reference counts an item when any of five samples names a cross-field gold work, so its figure is not a success@10.

System	Success (k of 10)	Quiz gold	Miss rate [Wilson]	Recall @10	MRR @10	Success @1	Leakage rate	$ per item
Single agent	3	2	0.70 [0.40, 0.89]	0.153	0.167	1	0.60	0.070
Prior-art skill	5	3	0.50 [0.24, 0.76]	0.270	0.263	1	0.70	0.130
Single agent, skeleton	4	3	0.60 [0.31, 0.83]	0.200	0.264	2	0.50	0.084
Memory reference	6	4	0.40 [0.17, 0.69]	n/a	n/a	n/a	n/a	none

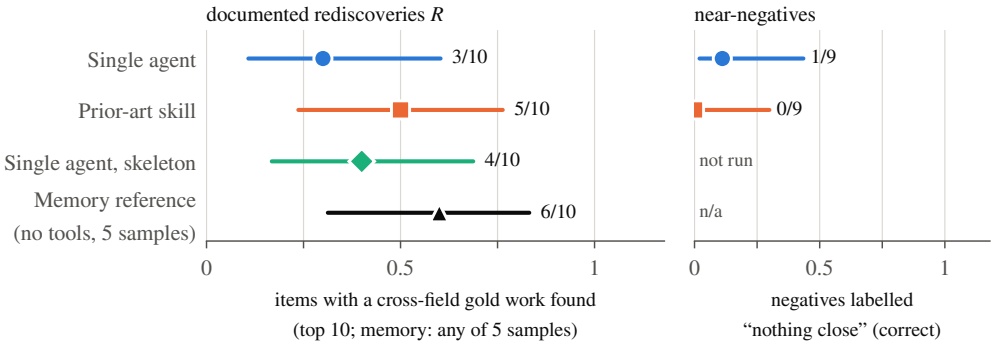


Figure 3. Retrieval on `R` and labelling of the near-negatives. Left: items with a cross-field gold work found in the top 10 (memory reference: any of five samples), with Wilson 95% intervals; gold includes the named works the author adjudicated. Right: near-negatives labelled `nothing close`, the correct label, for the two agents run on them.

condition, a neutral paraphrase of the claim with its domain stripped, finds 2 items that the text condition misses and misses 1 that it finds ($p = 1.0$). Of the 45, 66 and 57 cleaned candidates the agents return on `R`, 6, 13 and 8 are gold; the others are unjudged.

Verdicts. On `R` plus the 9 near-negatives, 19 items, the single agent has macro-F1 0.261 and the prior-art skill 0.198, because both call almost every negative related prior work: `nothing_close` for 1 of 9 and 0 of 9, on near-negatives with unchecked claim texts. "Already done" is predicted for 3 of 18 weaker items by the single agent and 1 of 18 by the prior-art skill, and the one item that is already done is called so by neither. The skeleton run has no negatives, so its macro-F1 of 0.600 on `R` alone is not comparable. The five scored runs cost $4.88.

8 Limitations

v0 was planned at about 80 items and ships with what is adjudicated: `R` has 10 items against a target of 30, so no confirmatory test runs, and `H` is empty. **Labels.** The pair labels come from one quiz over a pre-filled packet, answered in the author's presence by five annotators sharing three IDs, and agreement misses every threshold; Section 3.7 reports the anchoring check. A reader who needs independent human labels should treat them as one supervised reading. **Attribution.** One joint reading after the freeze decided 4 attribution pairs, for cf-0031 from a deposited list of 2 references that may be incomplete; Measurement A ran on the frozen `R`, without cf-0031 (Section 5.3). **Recall.** Three recalled items count only through named works read once each, and cf-0036's are cited by its source (Section 5.3). **Probes.** Two prompts are a narrow instrument: a miss is weak evidence



Figure 4. The 10 items of `R`: closed-book recall (left) and what each system found (right). Left, by model and prompt (frozen or cross-field, "cross"): blue marks a decided match that names a cross-field gold work in at least one of the five (frozen) or three (cross-field) samples; the last column marks the three items recalled only through a named work the author judged gold; cf-0044 is hatched where Opus 5.5 declined every sample, and its Haiku 4.5 cells hold only undecided loose matches. Right: items for which a system has a cross-field gold work in its top 10; markers follow Figure 3.

that a model has not memorised the link [20, 21], a hit does not show that a system used memory, and all probes are Claude models. **Scope.** The benchmark holds documented rediscoveries, which are exposed by construction, and says nothing about undocumented ones. Its sources are biased: declaration phrase search has low precision, and Kulikowski's corpus is computational. The other tools of Table 1 are not measured.

9 Ethics, Licences and AI Use

Five annotators took part, briefed in writing and with a kit in Russian: two women and three men aged 26 to 37, with graduate training in STEM, three with an MSc and two with a PhD, from universities in the United States, the United Kingdom and Kazakhstan. The author states that they were paid $12 to $14 per hour; the posted offer said $8 to $10. They appear in the data only as IDs a1 to a3, may be named in the acknowledgements if they wish, and keep item files confidential until release. The author reports that all five agreed in writing to take part and to have their answers released under an ID only. Items describe relations between published works, and the wording never accuses.

Code is MIT and data CC-BY-4.0; OpenAlex metadata is CC0, and Kulikowski's derived data and the Gradient Coupling catalogue are CC-BY-4.0 and attributed. No Semantic Scholar or Asta records or full texts are shipped.

Disclosure of interests. The author co-founded AgentMash, a company building coordination tools for AI coding agents, and wrote Hunch, an open-source (MIT) Claude Code agent shipped in AgentMash's agentmash lab plugin. Hunch checks whether a research idea was published before, including under another field's name, with relation labels like this benchmark's, so it is a checker of the kind measured here. It was planned as a system under test (protocol decision 10) and is not run in this version.

AI use. The annotators and the author made the decisions counted in Table 4. Claude proposed the quiz options and 33 candidate works, drafted claim texts, charts and evidence as pre-fill, screened the declarations and chose those in the batches from two blind readings, read the pilots' borderline matches and named works, generated the AI-origin items, decided the attribution of 4 pairs before the freeze under the author's delegation, ran the harness, wrote code and helped draft this paper. The annotators decided those 4 pairs again on 21 October. Codex transcribed or repackaged returns at the author's request. The author is responsible for all content of this paper. Scripts draw the figures from the result files.

10 Availability

A clean public repository, https://github.com/IskakovDamir/coelacanth-benchmark, holds the code (MIT), the protocol and its deviations log, the datasheet, the loader with its tests, the closed-book runner and scorer, the harness and scorer of measurement A and the one-command Kulikowski reproduction. The items with their gold, the closed-book answers, the run records with cost ledgers, the quiz returns with their provenance notes and the supplement ship in one password-protected archive with a canary string, as protocol decision 26 requires for items and gold. The archive is on OSF (DOI 10.17605/OSF.IO/672RJ), with mirrors on Hugging Face (DamirIskakov/coelacanth-v0) and Figshare (https://doi.org/10.6084/m9.figshare.34337127). The password, which keeps crawlers out and is not a secret, and the decryption command are in the datasheet, `eval/benchmark/DATASHEET.md`; the password moved there from a README, a logged deviation from decision 26.

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Appendices

These appendices reproduce the supplement that ships in the data archive (Section 10), without the passages that name gold works of benchmark items beyond the two Section 5.3 names (Yule 1925 for cf-0015, the PageRank papers for cf-0036), which protocol decision 26 keeps out of plain text.

A Pilots, supply and the decision on `H` (Sections 5.1 and 5.2)

A.1 Pilots

Three pilots preceded the v0 runs, on Claude-drafted claim texts and gold lists and with borderline matches judged by blind Claude readers under the author's delegation; the v0 runs replace them. The first (24 Sep) probed 20 items with three samples per model (180 calls, $2.34): no model named cross-field gold for 14 of the 20 items. The second (7 Oct) re-ran them with five frozen-prompt samples and the cross-field prompt (360 calls, $5.37): 9 of 20 items had no frozen-prompt sample naming cross-field gold, the cross-field prompt caught 8 of these 9 under a rule fixed before its first call, and 1 of 20 fails both probes (an item not in v0). What the cross-field prompt surfaces is mostly the method's origin in another field (statistics for population pharmacokinetics, linear prediction theory for kriging); under either prompt all 16 Kulikowski items were named. The declarations pilot (7 Oct; 323 calls, $5.07) probed 13 items, the 11 declarations and the 2 biology controls: 3 of 13 had no sample naming any gold work, all declarations, and the models named the declared earlier work for 8 of the 11 declarations.

A.2 Supply and what it implies for `H`

The declaration queries return about 823 works dated on or after 2026-07-01, which at round 2's rates is about 1 to 2 usable items, and the July filter overstates supply against the protocol's 2026-08-01 boundary; mined declarations therefore cannot feed `H-post`, and the AI-origin stratum is post-cutoff by construction and reported on its own. The pilots projected `H` at about 5 to 7 items (about 17 under the most generous reading), under the 20 that a test needs, so the protocol was updated on 9 Oct, before any human label, any system run on a benchmark item and the rules freeze: the union rule stays, `H`'s k of n comes first in every headline table, descriptively, and the memory reference is added. Gold stays as adjudicated: protocol decision 1, fixed on 6 Oct, makes every work a model names closed-book a candidate for its item, judged before `H` is computed; the option rejected on 9 Oct was a rule redefining gold from what the models named. The second probe stays as well, after it had caught 8 of 9 frozen-failed items. The paper therefore reports closed-book recall per item, model and prompt, checker results on `R` split by recall, `H` as k of n and recall by source. No test is run on `H` and no subset is called contamination-free. Recall differences between sources are confounded with popularity and are not read as a memorization effect, and undocumented rediscoveries are outside the scope.

B The v0 closed-book runs: bookkeeping, declines, borderline decisions and the named works (Section 5.3)

All 51 positives of b01 and b02 (32 human-origin rediscovery candidates, the two non-computational controls among them, 8 attributed controls, 10 AI-origin items, and one natural distractor that labelling found gold for) were probed on 10 and 11 October 2026 under both prompts, 1,224 planned samples: 955 new calls at $14.43 on 10 October, plus the pilot answers of the b01 items whose

claim text and date were unchanged (8 items still carry them). After the annotators' claim-text check, the 21 items whose text changed were probed again and their earlier answers set aside (550 rows); the re-probe stopped on a low credit balance, and the two R items it had left open (cf-0007, cf-0044) were completed afterwards with 18 calls for $0.34. A completion run on 2026-10-11 filled the cells the credit stop had left open (274 calls, $3.70, declines retried once). On b01 and b02, 1,215 of the 1,224 planned samples are done (valid, or declined twice): 1,183 valid and 32 Opus 5.5 samples declined twice. The 9 missing are Opus 5.5 cross-field samples on three AI-origin items, each declined once and not retried. The b01 and b02 calls number 1,501 for $21.43, re-probes included, and 197 answers are reused from the declarations pilot of 6 and 7 Oct. The two b03 items (cf-0068, cf-0069) were probed as well, 48 samples in 48 calls for $0.91, and are not counted; with them the run holds 53 items, 1,272 planned samples, 1,263 done and 1,549 calls for $22.34. Opus 5.5 declined 75 of its 454 counted answers on b01 and b02 (52 of 284 under the frozen prompt, 23 of 170 under the cross-field prompt), almost all on biology, medicine and immunology claims, and after the retries gave no valid sample on five items: cf-0006 (a control), cf-0044 (a Kulikowski twin on viral genome classification) and three AI-origin items. Those five cannot enter H; Sonnet 5.5 and Haiku 4.5 declined none. The 213 borderline matches of the first run were decided by a fixed rubric, 37 of them as naming the gold work. The author's latest account is that the paid annotators decided them; his earlier logged accounts differ (first split among the annotators and himself, later himself alone), and the export's decided_by value is one the page wrote automatically into every export, so it does not record who decided any entry. Decisions are keyed by item, model, prompt, sample and gold work, so 55 of them, on 7 of the 21 re-probed items, refer to answers the re-probe superseded and apply to the new answers by key (8 of them as naming the gold work); 27 loose matches of the re-probe were undecided by the count of the 12 October audit, 4 of them on cf-0007 and 3 on cf-0044; a recount of 21 October on the current answers, which also counts strict matches below 0.95 title overlap, finds four on cf-0044, all by Haiku 4.5, three under the frozen prompt and one under the cross-field prompt. None of this changes the status of an R item: cf-0007 is recalled by strict matches and cf-0044 leaves H through the declines.

Figure 2 shows the path from the items to H. Of the 11 provisional R items, 7 were recalled (some sample names a cross-field gold work) and 4 were not. cf-0031, a recalled item, is outside the frozen R because the attribution of its one pair was unknown at the freeze (Section 3.7), so the frozen R holds 10 items. Six are recalled directly (7 if an undecided loose match for cf-0044 is accepted). For three more, cf-0013, cf-0036 and cf-0046 (all Kulikowski twins), no model names a cross-field gold work; they are recalled only through works the models named outside the gold lists, which the author judged cross-field gold (below). The tenth, cf-0044, is in neither group, because Opus 5.5's safeguards declined every sample on its checked text and decision 6 keeps such an item out of H; Sonnet 5.5 and Haiku 4.5 gave valid samples and named no gold in a decided match, and the 5 works the models named for cf-0044 were never judged. If the declines were ignored, Haiku 4.5's four undecided loose matches read as misses and none of those 5 works rated gold, H would be 1 of 10 (Wilson 0.02 to 0.40). The annotators' reading of 21 October, a logged deviation, finds cf-0031's pair not cited (Section 3.7); with it R would hold 11 items, 10 of them recalled, and H would be 0 of 11 (Wilson 0.00 to 0.26). All declaration items in R are recalled (four of four in the frozen R, five of five with cf-0031), as the pilot suggested: a published declaration puts the link itself into the literature models are trained on. On the 10 items of R, the cross-field prompt raises the number of items whose cross-field gold a model names directly from 4 to 6 for Opus 5.5, from 4 to 6 for Sonnet 5.5 and from 2 to 4 for Haiku 4.5 (from 3 to 5 if undecided loose matches are accepted); under the frozen prompt alone 4 items are named by some model, and under either prompt 6 (5 and 7 with those matches). Sonnet 5.5 under the frozen prompt, the memory reference of Section 4, names a cross-field gold work for 4 of the 10 items (cf-0007, cf-0012, cf-0017, cf-0061). With the

named works the author judged gold, which Section 7 uses, it names one for 6 (Figure 4, left, shows closed-book recall on `R`).

Annotators judged the works the models named outside the gold lists for the three items for which no model named a cross-field gold work: 31 distinct works (cf-0013 6, cf-0036 16, cf-0046 9), one reading each, by hand on printed sheets under ID a1 (cf-0013, cf-0036) or a2 (cf-0046), each ID shared by two people; Codex transcribed the sheets. 14 were rated gold (any relation but `different`), and 6 of them cross-field gold. The author adjudicated all 14 on 11 Oct: 13 as the annotator read them, and one lowered from `equivalent` to `partial`. He confirmed every cross-field gold reading: 1 for cf-0013, 2 for cf-0036 and 3 for cf-0046. Each such work was named in a closed-book answer, so each of the three items counts as recalled. `H`, the `R` items that no model recalls under either probe, is therefore empty: 0 of 10 (Wilson 95% interval 0.00 to 0.28), and so are `H-any` and `H-post`. A sensitivity line added after the freeze: the two named works that make cf-0036 recalled, the PageRank papers, are cited by the item's own source, so if recall required a work the source does not cite, `H` would be 1 of 10 (Wilson 0.02 to 0.40). The closed-book control shows that every documented cross-field rediscovery in v0 that could be probed is recalled by at least one of three models, directly or through a work it names, so a retrieval score on `R` cannot be separated from memory. A re-decision of 30 random borderline entries by the author agreed with the first decisions on all 30; it is not counted as an independent second reading because one of his earlier logged accounts has him deciding the first run himself.

C Measurement A: systems, harness, metrics, statistics, budget and freezes (Section 4)

Systems. Two agent configurations are scored (Table 6): a single agent with web search, and the same agent with the prior-art skill [2], both Claude Code with the same model, effort, tools and caps; the single agent also runs in the skeleton condition, a logged deviation. The memory reference, Sonnet 5.5 without tools under the frozen closed-book prompt, is scored on `R` like a system, and works it names dated on or after the claim date count as parametric leakage. Kulikowski's fingerprint is reproduced on its own closed pool (Section 6) and is not scored on `R`. The protocol's memory reference is the first sample; the five samples name the same items. The protocol planned ten systems. The re-implemented INC-, OpenNovelty- and Idea-Catalyst-style checkers, Ai2 Theorizer, Owl and the author's own tools, Hunch and a reference pipeline, are not run in this version, because the API credit the author stated as remaining on 11 Oct ($10.87) did not cover them and no annotators remain to judge their pools; the paper says nothing about the other tools. Pooled judging is replaced by scoring against the gold already adjudicated: a retrieved work outside the gold counts as unjudged, so every retrieval figure is a lower bound.

Harness and metrics. The two agents share Sonnet 5.5 at high effort, 20 turns, 3 searches, 3 fetches, a $0.40 per-item cap and a 900-second timeout. CLI probes showed that the budget is checked only after a turn and that web tools call Haiku 4.5 internally, which we declare as an auxiliary model; an adversarial review of the harness confirmed 38 findings, all repaired before the freeze. An ambiguous native label maps to the one claiming less prior art, and unmappable outputs are counted. The primary metric is cross-field success@10, the share of items with a cross-field gold work in the top 10, reported as the miss rate with Wilson intervals; secondary metrics are cross-field recall@10, MRR@10 and success@1, and the same over all gold. Verdicts are scored by three-class macro-F1 (already done, related, nothing close) on `R` plus `N_built` (the 9 near-negatives), with the six-class confusion matrix descriptive; already-done precision and the false already-done rate are reported separately. Brier score and ECE (10 bins) are reported where a probability is given, and

Table 6. Systems in this version: the two agents and the memory reference are scored on `R`, and the Kulikowski fingerprint is reproduced on its own pool.

System	Access	Subset	Cost	Label mapping	Status
Single agent with web search	Claude Code, Sonnet 5.5	`R`, `N_built`	about $0.10 per item (pilot, 2 items)	strongest candidate relation	scored: `R` and `N_built` (text), `R` (skeleton)
Prior-art skill	Claude Code with an MIT prompt, hashed	`R`, `N_built`	about $0.16 per item (pilot, 2 items)	strongest candidate relation	scored: `R` and `N_built` (text)
Memory reference	Sonnet 5.5, no tools	`R`	none beyond the closed-book runs	named works only	scored on `R` from the closed-book answers (Section 5.3)
Kulikowski fingerprint + TF-IDF	reproduced package	`K`	free	ranks only	reproduced (Section 6)

leakage rate and cost per item for the two agents.

Statistics. `R` holds 10 items, below the 20 that decision 23 requires of a confirmatory test, so every comparison is descriptive: k of n with Wilson intervals, and exact McNemar comparisons of the two agents on cross-field success@10, reported as exploratory with their discordant counts. `H` and every subset under 20 items are descriptive only.

Budget and freezes. The scored runs are paid from the $10.87 of API credit that remained on 11 Oct; the cost pilots on two items outside v0 cost $0.1988 and $0.3267. The rules freeze (tag `protocol-v1.0`, 11 October 2026) was planned before the quiz labelling and the closed-book runs and came after them; the changes made in between are logged as updates in section 14 of the protocol, and neither agent had run on a benchmark item (the memory reference reuses closed-book answers collected before the freeze). The freeze covers the rules, each agent configuration (hashed before its first scored run) and the items and splits; later changes are logged as deviations, among them the single agent's skeleton run, decided after the first scored runs had been looked at, and the correction of the gold in Section 7.

D Kulikowski reproduction and the closed-pool protocol (Section 6)

We reproduced the TF-IDF rows of Kulikowski's benchmark from his package v1.0.0 with one command (2,181 files verified, about 40 seconds, no model call). Every fingerprint number reproduces exactly (Table 7); the abstract rows differ by at most one query (P@1 52 against 51 of 92, AP 0.223 against 0.222), because 5 abstracts are not redistributable and were rebuilt from OpenAlex.

Three points matter for comparisons with this baseline. First, the package has two TF-IDF settings: the headline default (unigrams and bigrams, sublinear term frequency, English stop words), behind every AP in Kulikowski's paper, and scikit-learn's plain unigram vectorizer, behind his whole-fingerprint P@1 of 0.826. His whole-fingerprint AP of 0.557 and P@1 of 0.826 therefore come from different settings: under the unigram setting AP is 0.441, and under the headline setting P@1 is 0.870. The pair "AP 0.557, P@1 0.82" in this project's early planning notes also mixes two representations, since P@1 0.82 is the skeleton's. Comparisons use the unigram setting with the headline setting as a sensitivity line. Second, the pool is closed: a system that searches open literature can retrieve correct earlier works outside the 109 papers, which the pool's judgments count as misses, so the closed-pool protocol has every system rank only the pool's cross-field candidates

Table 7. Kulikowski's closed pool (109 papers, 92 queries, 5,812 cross-field pairs, 210 twins): paper against reproduction, and our harness's closed-pool rankings.

Metric	Representation	Paper	Reproduced	Our harness
AP	whole fingerprint + TF-IDF (headline settings)	0.557 [0.407, 0.720]	0.557 [0.407, 0.720]	0.557
AP	whole fingerprint + unigram TF-IDF	not printed		0.441
P@1 / MRR	skeleton + TF-IDF	0.82 / 0.86	0.815 (75 of 92) / 0.860	
P@1 / MRR	whole fingerprint + unigram TF-IDF	0.826	0.826 / 0.870	0.826 / 0.870
P@1 / MRR	whole fingerprint + headline TF-IDF settings	not printed	0.870 (our diagnostic)	0.870 / 0.912

for each query, scored by P@1 and MRR per query and by AP over the 5,812 pairs. Third, the families are well-known imports: in the first pilot 6 of 16 Kulikowski items were named closed-book by at least one model, and in the second probe all 16 were under either prompt (Claude readings). No other system is run in the closed pool in this version; the closed-pool protocol stays available to others.

E Pooled judging as specified in the protocol (Section 4)

The protocol specifies TREC-style pooled judging: the top 10 candidates of every system, the annotators' own candidates and the works named closed-book form each item's pool, judged in two stages by the item's annotator blind to system and rank, with the author adjudicating every candidate rated `similar_mechanism` or stronger. This version does not carry it out: no annotator remains to judge pools, so system output is scored against the gold already adjudicated and a retrieved work outside the gold counts as unjudged (Section 4).